1.04 Complex Numbers: Roots of Equations

Worked Solutions

1 a)

Three real roots

One real root and two complex roots

1 b)

Four real roots

Two real roots and two complex roots

Four complex roots

1 c)

Five real roots

Three real roots and two complex roots

One real root and four complex roots

2)

Two roots are $1 + 5i$ and $1 - 5i$ which gives the quadratic: $z^2 - 2z + 26$

$$z^3 - 4z^2 + 30z - 52 = (z^2 - 2z + 26)(z - 2)$$

So the three roots are $1 + 5i$, $1 - 5i$ and 2

3)

Two roots are $3 - 5i$ and $3 + 5i$ which gives the quadratic: $z^2 - 6z + 34$

$$-z^3 + 11z^2 - 64z + 170 = (z^2 - 6z + 34)(-z + 5)$$

So the three roots are $3 - 5i$, $3 + 5i$ and 5

4)

The roots are 4, $2 - 8i$ and $2 + 8i$

$$(z - (2 - 8i))(z - (2 + 8i)) = z^2 - 4z + 68$$
$$(z - 4)(z^2 - 4z + 68) = z^3 - 4z^2 + 68z - 4z^2 + 16z - 272$$

So the polynomial is $z^3 - 8z^2 + 84z - 272$

$\therefore a = -8$, $b = 84$ and $c = -272$

5)

The roots are -3, $1 + 3i$ and $1 - 3i$

$$(z - (1 + 3i))(z - (1 - 3i)) = z^2 - 2z + 10$$
$$(z + 3)(z^2 - 2z + 10) = z^3 - 2z^2 + 10z + 3z^2 - 6z + 30$$

So the equation is $z^3 + z^2 + 4z + 30 = 0$

6 a)

$1 - 4i$ and 1

6 b)

$$\begin{aligned}&(z-1)(z-(1+4i))(z-(1-4i))\\&=(z-1)(z^2-2z+17)\\&=z^3-2z^2+17z-z^2+2z-17\\&=z^3-3z^2+19z-17\end{aligned}$$

7)

The roots are $1+i$, $1-i$, -2 and 3

$$(z-(1+i))(z-(1-i)) = z^2-2z+2$$
$$(z+2)(z-3) = z^2-z-6$$
$$\begin{aligned}&(z^2-2z+2)(z^2-z-6)\\&=z^4-2z^3+2z^2-z^3+2z^2-2z-6z^2+12z-12\\&=z^4-3z^3-2z^2+10z-12\end{aligned}$$

$\therefore a=-3$, $b=-2$, $c=10$, $d=-12$

8)

The roots are $1-5i$, $1+5i$, $2+i$ and $2-i$

$$(z-(1+5i))(z-(1-5i)) = z^2-2z+26$$
$$(z-(2+i))(z-(2-i)) = z^2-4z+5$$
$$(z^2-2z+26)(z^2-4z+5) = z^4-6z^3+39z^2-114z+130$$

So the equation is $\mathrm{f}(z) = z^4-6z^3+39z^2-114z+130$

$a=1$, $b=-6$, $c=39$, $d=130$

9)

The roots are $2+i$, $2-i$, $4-3i$ and $4+3i$

$$(z-(2+i))(z-(2-i)) = z^2-4z+5$$
$$(z-(4-3i))(z-(4+3i)) = z^2-8z+25$$
$$\begin{aligned}&(z^2-4z+5)(z^2-8z+25)\\&=z^4-8z^3+25z^2-4z^3+32z^2-100z+5z^2-40z+125\\&=z^4-12z^3+62z^2-140z+125\end{aligned}$$

$\therefore a=-12$, $b=62$, $c=-140$, $d=125$

10)

One root is 9 so that $(z-9)$ is a factor

Therefore by inspection:

$$z^3-13z^2+56z-180 = (z-9)(z^2-4z+20)$$

$z^2-4z+20=0$ has roots $z=\frac{4\pm\sqrt{16-80}}{2}=2\pm4i$

So the three roots are 9, $2+4i$ and $2-4i$

11)

$(z-4)$ is a factor

Therefore by inspection:

$$z^3 - 10z^2 + 34z - 40 = (z-4)(z^2 - 6z + 10)$$

$z^2 - 6z + 10$ has roots $z = \frac{6 \pm \sqrt{36-40}}{2} = 3 \pm i$

So the three roots are 4, $3+i$ and $3-i$

12)

$(z+3)$ is a factor so -3 is a solution

Therefore by inspection:

$$z^3 + z^2 + 11z + 51 = (z+3)(z^2 - 2z + 17)$$

$z^2 - 2z + 17 = 0$ has solutions $z = \frac{2 \pm \sqrt{4-68}}{2} = 1 \pm 4i$

So the three solutions are -3, $1+4i$ and $1-4i$

13)

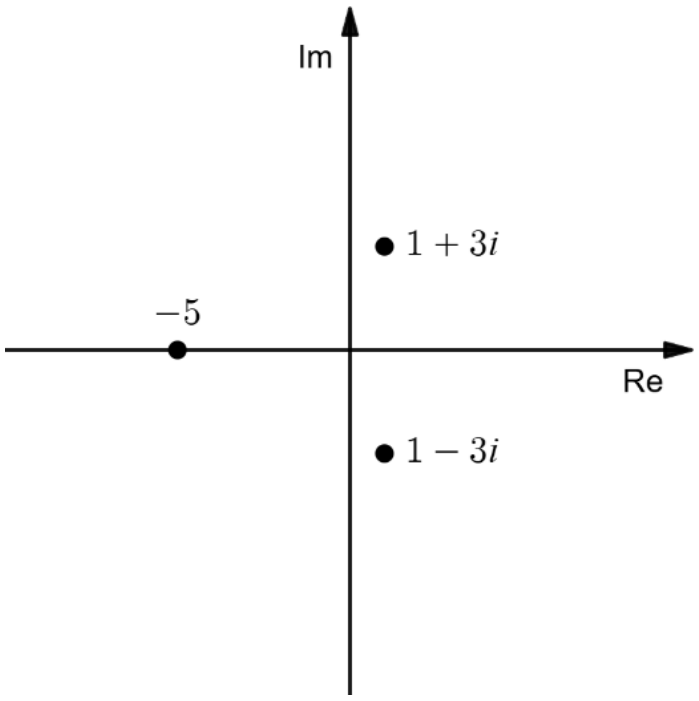
Two of the roots are $1-3i$ and $1+3i$

$$(z-(1-3i))(z-(1+3i)) = z^2 - 2z + 10$$

By inspection:

$$(z^2 - 2z + 10)(z+5) = z^3 + 3z^2 + 50$$

So the three roots are -5, $1-3i$ and $1+3i$



Area $= \frac{1}{2} \times 6 \times 6 = 18$ units^2

14)

One of the roots is 3

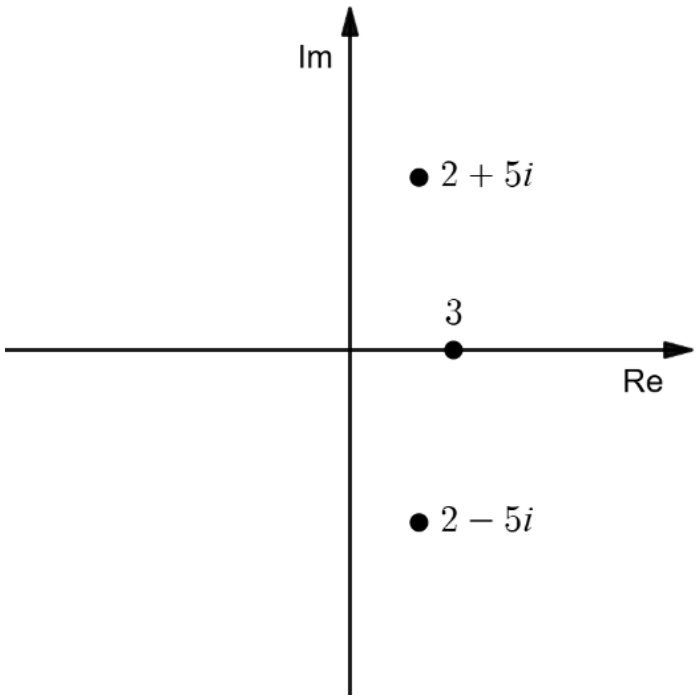
By inspection:

$$(z-3)(z^2-4z+29) = z^3-7z^2+41z-87$$

And solving $z^2-4z+29=0$ gives

$$z^2-4z+29=0$$
$$(z-2)^2-4=-29 \Rightarrow z-2=\pm\sqrt{-25} \Rightarrow z=2\pm 5i$$

So the three roots are 3, $2+5i$ and $2-5i$



Area $= \frac{1}{2} \times 1 \times 10 = 5$ units^2

15 a)

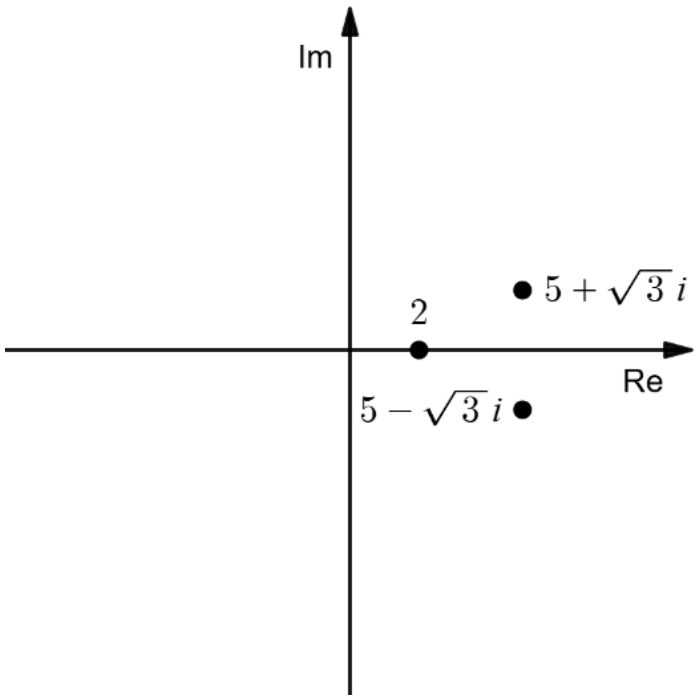
Two of the roots are $5-\sqrt{3}i$ and $5+\sqrt{3}i$

$$(z-(5-\sqrt{3}i))(z-(5+\sqrt{3}i)) = z^2-10z+28$$

By inspection:

$$(z^2-10z+28)(z-2) = z^3-12z^2+48z-56$$

So the three roots are 2, $5-\sqrt{3}i$ and $5+\sqrt{3}i$



Area $= \frac{1}{2} \times 2\sqrt{3} \times 3 = 3\sqrt{3}$ units^2

15 b)

Calculate lengths of sides:

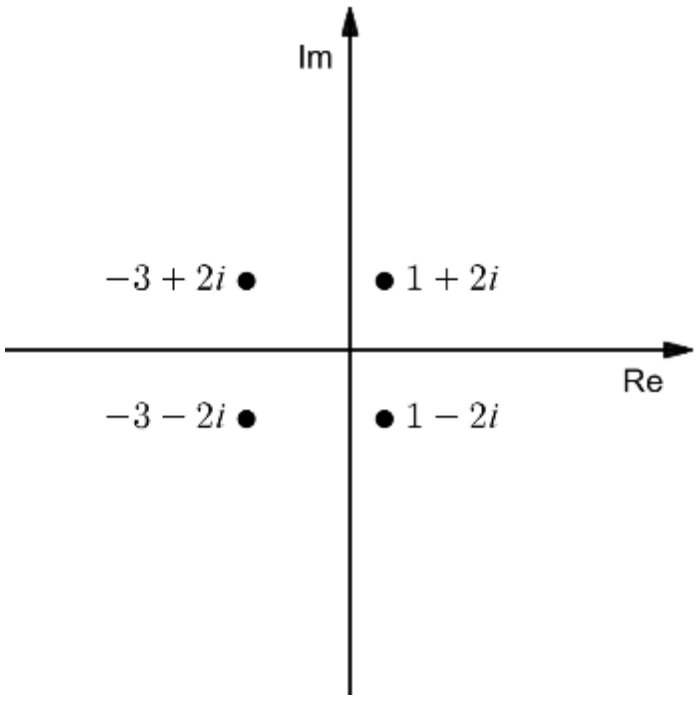
Between $5+\sqrt{3}i$ and $5-\sqrt{3}i$ it is $2\sqrt{3}$

Between 2 and $5+\sqrt{3}i$ it is $\sqrt{3^2+(\sqrt{3})^2} = \sqrt{12} = 2\sqrt{3}$

Between 2 and $5-\sqrt{3}i$ it is $\sqrt{3^2+(-\sqrt{3})^2} = \sqrt{12} = 2\sqrt{3}$

Since all the lengths are the same it must be an equilateral triangle.

16 a)



16 b)

$$\begin{aligned}
&(z-(1+2i))(z-(1-2i))(z-(-3+2i))(z-(-3-2i)) \\
&= (z^2-2z+5)(z^2+6z+13) \\
&= z^4+6z^3+13z^2-2z^3-12z^2-26z+5z^2+30z+65 \\
&= z^4+4z^3+26z^2+4z+65 \\
\therefore \mathrm{f}(z) &= z^4+4z^3+26z^2+4z+65
\end{aligned}$$

17 a)

$3+5i$, $3-5i$ and 3

17 b)

$$\begin{aligned}
&(z-(3+5i))(z-(3-5i))(z-3)(z-3) \\
&= (z^2-6z+34)(z^2-6z+9) \\
&= z^4-6z^3+9z^2-6z^3+36z^2-54z+34z^2-204z+306 \\
&= z^4-12z^3+79z^2-258z+306 \\
\therefore \mathrm{f}(z) &= z^4-12z^3+79z^2-258z+306
\end{aligned}$$

18 a)

$$\begin{aligned}
(2+4i)^3 &= 2^3+3\times 2^2\times 4i+3\times 2\times (4i)^2+\times(4i)^3 \\
&= 8+48i-96-64i \\
&= -88-16i
\end{aligned}$$

18 b)

$$\begin{aligned}
&(2+4i)^3-7(2+4i)^2+32(2+4i)-60 \\
&= -88-16i-7(-12+16i)+64+128i-60 \\
&= 0
\end{aligned}$$

$\therefore$ by the factor theorem, $2+4i$ is a root of $\mathrm{f}(z)=0$

18 c)

Two of the roots are $2+4i$ and $2-4i$

$$(z-(2+4i))(z-(2-4i)) = z^2-4z+20$$

By inspection:

$$z^3-7z^2+32z-60 = (z^2-4z+20)(z-3)$$

19 a)

$$\begin{aligned}
&(1+i)^4-(1+i)^3-6(1+i)^2+14(1+i)-12 \\
&= 2i\times 2i-2i(1+i)-6\times 2i+14+14i-12 \\
&= -4-2i+2-12i+14+14i-12 \\
&= 0
\end{aligned}$$

$\therefore 1+i$ is a root.

19 b)

Two of the roots are $1+i$ and $1-i$

$$(z-(1+i))(z-(1-i)) = z^2-2z+2$$

	z^2	$+z$	-6
z^2	z^4	$+z^3$	$-6z^2$
$-2z$	$-2z^3$	$-2z^2$	$+12z$
$+2$	$+2z^2$	$+2z$	-12

And

$z^2+z-6=(z-2)(z+3)=0$ has solutions $z=2$ and $z=-3$

$\therefore$ the roots of $z^4-z^3-6z^2+14z-12=0$ are 2, -3, $1+i$ and $1-i$

20)

Two of the roots are $2-7i$ and $2+7i$

$$(z-(2-7i))(z-(2+7i)) = z^2-4z+53$$

	z^2	$-2z$	$+10$
z^2	z^4	$-2z^3$	$+10z^2$
$-4z$	$-4z^3$	$+8z^2$	$-40z$
$+53$	$+53z^2$	$-106z$	$+530$

And

$z^2-2z+10=0$ has solutions $z=1-3i$ and $z=1+3i$

$\therefore$ the roots of $z^4-6z^3+71z^2-146z+530=0$ are $2-7i$, $2+7i$, $1-3i$ and $1+3i$

21 a)

Two of the solutions are $1+3i$ and $1-3i$

$$(z-(1+3i))(z-(1-3i)) = z^2-2z+10$$

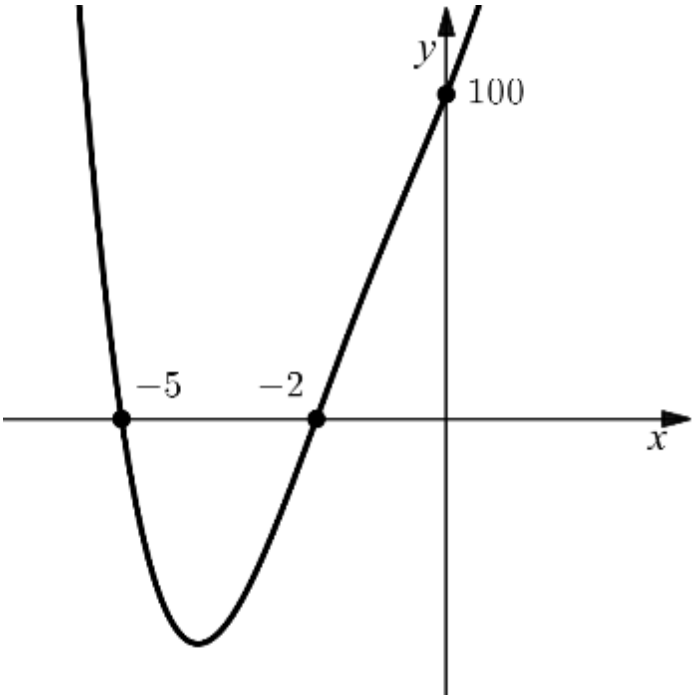
	z^2	$+7z$	$+10$
z^2	z^4	$+7z^3$	$+10z^2$
$-2z$	$-2z^3$	$-14z^2$	$-20x$
$+10$	$+10z^2$	$+70z$	$+100$

And

$z^2+7z+10=(z+5)(z+2)=0$ has solutions $z=-5$ and $z=-2$

$\therefore$ the solutions of $z^4+5z^3+6z^2+50z+100=0$ are -5, -2, $1+3i$ and $1-3i$

21 b)



22 a)

$$\begin{aligned}&(1+2i)^4+4(1+2i)^3+3(1+2i)^2+10(1+2i)+50\\&=(-3+4i)^2+4(-3+4i)(1+2i)+3(-3+4i)+10(1+2i)+50\\&=-7-24i-44-8i-9+12i+10+20i+50\\&=0\end{aligned}$$

$\therefore 1+2i$ is a root

22 b)

Two of the solutions are $1+2i$ and $1-2i$

$$(z-(1+2i))(z-(1-2i))=z^2-2z+5$$

	z^2	$+6z$	$+10$
z^2	z^4	$+6z^3$	$+10z^2$
$-2z$	$-2z^3$	$-12z^2$	$-20z$
$+5$	$+5z^2$	$+30z$	$+50z$

And

$z^2+6z+10=0$ has solutions $z=-3+i$ and $z=-3-i$

$\therefore$ the roots of $z^4+4z^3+3z^2+10z+50=0$ are $1+2i$, $1-2i$, $-3+i$ and $-3-i$

22 c)

All of the solutions of $\mathrm{f}(x)=0$ are complex and so the graph of $y=\mathrm{f}(x)$ will not intersect the x-axis.

23 a)

Using factor theorem:

$$\begin{aligned} \mathrm{f}(z) &= z^3 - 13z^2 + 56z - 78 \\ \mathrm{f}(1) &= 1 - 13 + 56 - 78 \neq 0 \\ \mathrm{f}(2) &= 8 - 52 + 112 - 78 \neq 0 \\ \mathrm{f}(3) &= 27 - 117 + 168 - 78 = 0 \ \therefore (z-3) \text{ is a factor} \\ z^3 - 13z^2 + 56z - 78 &= (z-3)(z^2 - 10z + 26) \\ z &= \frac{10 \pm \sqrt{100 - 104}}{2} \end{aligned}$$

So the solutions are $z = 3$, $z = 5 - i$ and $z = 5 + i$

23 b)

Using factor theorem:

$$\begin{aligned} \mathrm{f}(z) &= z^3 - 6z^2 + 15z - 14 \\ \mathrm{f}(1) &= 1 - 6 + 15 - 14 \neq 0 \\ \mathrm{f}(2) &= 8 - 24 + 30 - 14 = 0 \ \therefore (z-2) \text{ is a factor} \\ z^3 - 6z^2 + 15z - 14 &= (z-2)(z^2 - 4z + 7) \\ z &= \frac{4 \pm \sqrt{16 - 28}}{2} \end{aligned}$$

So the solutions are $z = 2$, $z = 2 - \sqrt{3}i$ and $z = 2 + \sqrt{3}i$

23 c)

Using factor theorem:

$$\begin{aligned} \mathrm{f}(z) &= z^3 - 2z^2 - 2z + 12 \\ \mathrm{f}(1) &= 1 - 2 - 2 + 12 \neq 0 \\ \mathrm{f}(2) &= 8 - 8 - 4 + 12 \neq 0 \\ \mathrm{f}(-2) &= -8 - 8 + 4 + 12 \ \therefore (z+2) \text{ is a factor} \\ z^3 - 2z^2 - 2z + 12 &= (z+2)(z^2 - 4z + 6) \\ z &= \frac{4 \pm \sqrt{16 - 24}}{2} \end{aligned}$$

So the solutions are $z = -2$, $z = 2 - \sqrt{2}i$ and $z = 2 + \sqrt{2}i$

23 d)

Using factor theorem:

$$\begin{aligned} \mathrm{f}(z) &= 3z^3 - 20z^2 + 54z - 28 \\ \mathrm{f}(1) &= 3 - 20 + 54 - 28 \neq 0 \\ \mathrm{f}\left(\frac{1}{3}\right) &= \frac{1}{9} - \frac{20}{9} + 18 - 28 \neq 0 \\ \mathrm{f}(2) &= 24 - 80 + 108 - 28 \neq 0 \\ \mathrm{f}\left(\frac{2}{3}\right) &= \frac{8}{9} - \frac{80}{9} + 36 - 28 = 0 \ \therefore (3z-2) \text{ is a factor} \\ 3z^3 - 20z^2 + 54z - 28 &= (3z-2)(z^2 - 6z + 14) \\ z &= \frac{6 \pm \sqrt{36 - 56}}{2} \end{aligned}$$

So the solutions are $z = \frac{2}{3}$, $z = 3 - \sqrt{5}i$ and $z = 3 + \sqrt{5}i$

24 a)

Since complex roots for polynomials with real coefficients are conjugate pairs, the other roots must be $1 - 4i$ and 1 (since they have to be in a straight line)

$\therefore 1 + 4i$, $1 - 4i$ and 1

24 b)

$$\begin{aligned}
&(z - (1 - 4i))(z - (1 + 4i))(z - 1) \\
&= (z^2 - 2z + 17)(z - 1) \\
&= z^3 - 2z^2 + 17z - z^2 + 2z - 17 \\
&= z^3 - 3z^2 + 19z - 17 \\
\therefore \mathrm{f}(z) &= z^3 - 3z^2 + 19z - 17
\end{aligned}$$

25 a)

Since complex roots for polynomials with real coefficients are conjugate pairs, the roots must be $2 - 3i$, $2 + 2i$ and 2 (since they have to be in a straight line)

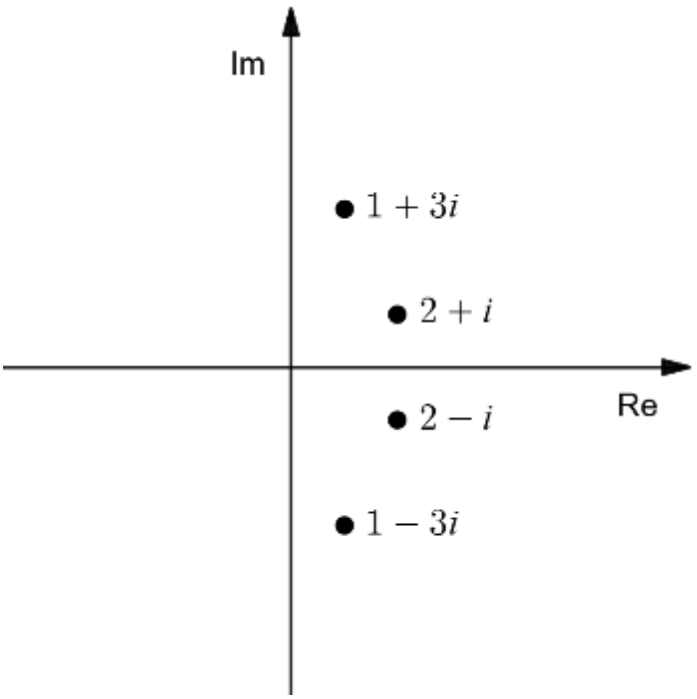
$\therefore 2 - 3i$, $2 + 3i$, $2 + 2i$, $2 - 2i$ and 2

25 b)

$$\begin{aligned}
&(z - (2 - 3i))(z - (2 + 3i))(z - (2 + 2i))(z - (2 - 2i))(z - 2) \\
&= (z^2 - 4z + 13)(z^2 - 4z + 8)(z - 2) \\
&= (z^4 - 4z^3 + 8z^2 - 4z^3 + 16z^2 - 32z + 13z^2 - 52z + 104)(z - 2) \\
&= (z^4 - 8z^3 + 37z^2 - 84z + 104)(z - 2) \\
&= z^5 - 8z^4 + 37z^3 - 84z^2 + 104z - 2z^4 + 16z^3 - 74z^2 + 168z - 208 \\
&= z^5 - 10z^4 + 53z^3 - 158z^2 + 272z - 208 \\
\therefore \mathrm{f}(z) &= z^5 - 10z^4 + 53z^3 - 158z^2 + 272z - 208
\end{aligned}$$

26 a)

The roots of the quartic are $2 + i$, $2 - i$, $1 - 3i$ and $1 + 3i$



26 b)

Area is $\frac{1}{2} \times 1 \times (2+6) = 4$ units^2

26 c)

$$\begin{aligned}(z-(2+i))(z-(2-i)) &= z^2-4z+5\\ (z-(1-3i))(z-(1+3i)) &= z^2-2z+10\\ (z^2-4z+5)(z^2-2z+10) &= z^4-2z^3+10z^2-4z^3+8z^2-40z+5z^2-10z+50\\ &= z^4-6z^3+23z^2-50z+50\end{aligned}$$

$\therefore a=-6$, $b=23$, $c=-50$ and $d=50$

27 a)

$$(1+2i)(1+2i) = 1+4i-4 = -3+4i$$

27 b)

$$\begin{aligned}&(z-(1+2i))(z-(1-2i))(z-(-3+4i))(z-(-3-4i))\\ &= (z^2-2z+5)(z^2+6z+25)\\ &= z^4+6z^3+25z^2-2z^3-12z^2-50z+5z^2+30z+125\\ &= z^4+4z^3+18z^2-20z+125\end{aligned}$$

28 a)

Since $1-2i$ is a root $1+2i$ is also a root

$$(z-(1-2i))(z-(1+2i)) = z^2-2z+5$$

$\therefore$ by inspection

$$z^3 - 4z^2 + kz - 10 = (z-2)(z^2 - 2z + 5)$$

So that the roots are $1 - 2i$, $1 + 2i$ and 2

28 b)

$$(z-2)(z^2 - 2z + 5) = z^3 - 4z^2 + 9z - 10$$

$\therefore k = 9$

29 a)

Let $\mathrm{f}(z) = z^3 + kz^2 - 5z - 12$

Since $z - 4$ is a factor:

$$\mathrm{f}(4) = 4^3 + k(4)^2 - 5(4) - 12 = 0$$
$$64 + 16k - 20 - 12 = 0$$
$$\therefore k = -2$$

29 b)

$$z^3 - 2z^2 - 5z - 12 = (z-4)(z^2 + 2z + 3)$$

So that the roots are 4, $-1 - \sqrt{2}i$ and $-1 + \sqrt{2}i$

30 a)

Since $3 - i$ is a root, then $3 + i$ is also a root:

$$(z - (3-i))(z - (3+i)) = z^2 - 6z + 10$$

	$2z$	-7
z^2	$2z^3$	$-7z^2$
$-6z$	$-12z^2$	$+42z$
$+10$	$+20z$	-70

So that $20 + 42 = p$ and $q = -70$

$\therefore p = 62$ and $q = -70$

30 b)

By inspection:

$$z^3 - 19z^2 + 62z - 70 = (z^2 - 6z + 10)(2z - 7)$$

So that the roots are $\frac{7}{2}$, $3 - i$ and $3 + i$

31 a)

Two of the solutions are $2 - i$ and $2 + i$

$$(z - (2-i))(z - (2+i)) = z^2 - 4z + 5$$
$$z^4 - 6z^3 + pz^2 + qz + 20 = (z^2 - 4z + 5)(z^2 - 2z + 4)$$

31 b)

$$p = 4 + 8 + 5 = 17$$
$$q = -16 - 10 = -26$$

32 a)

Since $2i$ is a root, then $-2i$ is also a root:

$$(z - 2i)(z - -2i) = z^2 + 4$$

	z^2	$-3z$	-10
z^2	z^4	$-3z^3$	$-10z^2$
$+0z$	$+0z^3$	$-0z^2$	$-0z$
$+4$	$+4z^2$	$-12z$	-40

$\therefore p = -12$ and $q = -40$

32 b)

$z^2 - 3z - 10 = 0 = (z + 2)(z - 5)$ has roots $z = -2$ and $z = 5$

The roots are $2i$, $-2i$, -2 and 5

33 a)

Let the root be ai for some real value a

Then

$$(ai)^4 - 6(ai)^3 + 2(ai)^2 - 54(ai) - 63 = 0$$
$$a^4 + 6a^3 i - 2a^2 - 54ai - 63 = 0$$
$$(a^4 - 2a^2 - 63) + (6a^3 - 54a)i = 0$$

Real part: $a^4 - 2a^2 - 63 = 0 \Rightarrow (a^2 - 3)(a^2 + 7) = 0$ so $a = \pm 3$ or $a = \pm\sqrt{7}i$

Since a is real $a = \pm 3$

So the purely imaginary roots are $3i$ and $-3i$

33 b)

$$(z - 3i)(z + 3i) = z^2 + 9$$
$$z^4 - 6z^3 + 2z^2 - 54z - 63 = 0$$
$$(z^2 + 9)(z^2 - 6z - 7) = 0$$
$$(z^2 + 9)(z - 7)(z + 1) = 0$$

So the roots are

$z = 7$, $z = -1$, $z = 3i$ and $z = -3i$

34)

Since $3 - 2i$ is a root, then $3 + 2i$ is also a root:

$$(z - (3 - 2i))(z - (3 + 2i)) = z^2 - 6z + 13$$

By inspection:

$$\begin{aligned} z^4 + az^3 + bz^2 + az + 13 &= (z^2 - 6z + 13)(z^2 + Pz + 1) \\ &= z^4 + Pz^3 + z^2 - 6z^3 - 6Pz^2 - 6z + z^2 - 6z + 13 \\ &= z^4 + (P-6)z^3 + (2-6P)z^2 - 6z + 13 \end{aligned}$$

Comparing linear terms: $a = -6$

Comparing cubic terms: $P - 6 = -6$ so $P = 0$

So

$$z^4 + az^3 + bz^2 + az + 13 = (z^2 - 6z + 13)(z^2 + 1)$$

35 a)

Since $-1 + \frac{\sqrt{3}}{2}i$ is a root, $-1 - \frac{\sqrt{3}}{2}i$ is also a root:

$$4\left(z - \left(-1 + \frac{\sqrt{3}}{2}i\right)\right)\left(z - \left(-1 - \frac{\sqrt{3}}{2}i\right)\right) = 4\left(z^2 + 2z + \frac{7}{4}\right) = 4z^2 + 8z + 7$$

$$4z^4 + 16z^3 + 51z^2 + kz + 49 = (4z^2 + 8z + 7)(z^2 + 2z + 7)$$

So $-1 + \sqrt{6}i$ and $-1 - \sqrt{6}i$ are also roots.

Therefore, the roots are $-1 + \frac{\sqrt{3}}{2}i$, $-1 - \frac{\sqrt{3}}{2}i$, $-1 + \sqrt{6}i$ and $-1 - \sqrt{6}i$

35 b)

$$k = 56 + 14 = 70$$

36 a)

$$(2 - i)(2 - i) = 4 - 4i - 1 = 3 - 4i$$

36 b)

$$(3 - 4i)(2 - i) = 6 - 8i - 3i - 4 = 2 - 11i$$

36 c)

$$\begin{aligned} p(2 - 11i) + 5(3 - 4i) + 12(2 - i) + q &= 0 \\ 2p - 11pi + 15 - 20i + 24 - 12i + q &= 0 \\ 2p + 39 + q + (-23p - 32)i &= 0 \end{aligned}$$

Imaginary part: $-23p - 32 = 0$ so $p = -\frac{32}{11}$

Real part: $2 \times \frac{20}{23} + 39 + q = 0$ so $q = -\frac{365}{11}$

36 d)

$$-\frac{32}{11}z^3 + 5z^2 + 12z - \frac{365}{11} = 0$$
$$32z^3 - 55z^2 - 132z + 365 = 0$$

Since we know $p\alpha^3 + 5\alpha^2 + 12\alpha + q = 0$ has solution $\alpha = 2 - i$ for these values of p and q and that the coefficients are real two of the roots are $2 - i$ and $2 + i$

$$(z - (2 - i))(z - (2 + i)) = z^2 - 4z + 5$$
$$32z^3 - 55z^2 - 132z + 365 = (z^2 - 4z + 5)(32z + 73)$$

So the solutions are $z = 2 - i$, $z = 2 + i$ and $z = -\frac{73}{32}$

37 a)

$z^2 - 3z + 9 = 0$ has roots $z = \frac{3}{2} \pm \frac{3\sqrt{3}}{2}i$

$$r = \sqrt{\left(\frac{3}{2}\right)^2 + \left(\frac{3\sqrt{3}}{2}\right)^2} = 3$$
$$\theta = \tan^{-1}\left(\frac{\frac{3\sqrt{3}}{2}}{\frac{3}{2}}\right) = \frac{\pi}{3}$$

So in modulus-argument form the roots are:

$$3\left(\cos\frac{\pi}{3} + i\sin\frac{\pi}{3}\right) \text{ and } 3\left(\cos\frac{\pi}{3} - i\sin\frac{\pi}{3}\right)$$

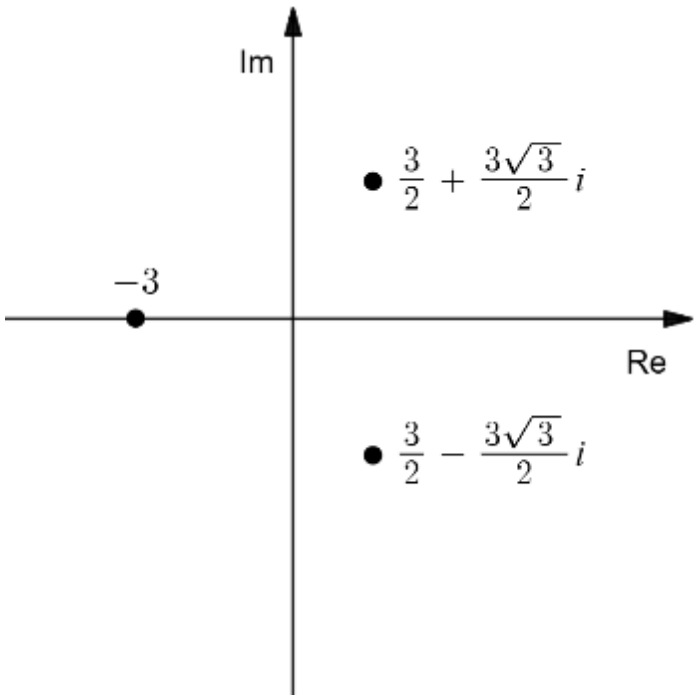
37 b)

$$z^3 + k = (z^2 - 3z + 9)(z + 3)$$

So $k = 27$

The final root is $z = -3$

37 c)



38 a)

By inspection,

$$1 + k - k - 1 = 0$$

$\therefore z = 1$ is a root of the equation.

38 b)

	z^2	$(1+k)z$	$+1$
z	z^3	$(1+k)z^2$	$+z$
-1	$-z^2$	$-(1+k)z$	-1

$$\therefore z^3 + kz^2 - kz - 1 = (z-1)(z^2 + (1+k)z + 1)$$

For $z^2 + (1+k)z + 1$ to have complex roots $b^2 - 4ac < 0$

$$\begin{aligned}(1+k)^2 - 4 \times 1 \times 1 &< 0 \\ k^2 + 2k - 3 &< 0 \\ (k+3)(k-1) &< 0 \\ \therefore -3 < k &< 1\end{aligned}$$

39 a)

Two of the roots are $5 + 2i$ and $5 - 2i$

$$(z - (5+2i))(z - (5-2i)) = z^2 - 10z + 29$$

$\therefore z^2 - 10z + 29$ is a factor of f(x)

39 b)

	z^4	$+0z^3$	$+0z^2$	$+0z$	-16
z^2	z^6	$+0z^5$	$+0z^4$	$+0z^3$	$-16z^2$
$-10z$	$-10z^5$	$-0z^4$	$-0z^3$	$-0z^2$	$+160z$
$+29$	$+29z^4$	$+0z^3$	$+0z^2$	$+0z$	-464

So $z^4 - 16$ is also a factor

By inspection:

$z^4 - 16 = 0$ has solutions 2, -2, $2i$ and $-2i$

So all the solutions are $5 + 2i$, $5 - 2i$, 2, -2, $2i$ and $-2i$

40)

Two of the solutions are $4 - 7i$ and $4 + 7i$

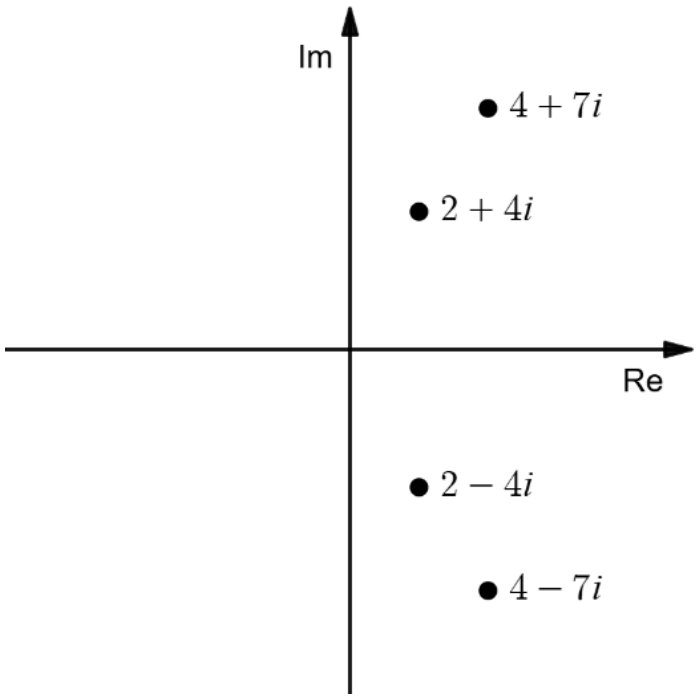
$$(z - (4 - 7i))(z - (4 + 7i)) = z^2 - 8z + 65$$

	z^2	$-4z$	$+20$
z^2	z^4	$-4z^3$	$20z^2$
$-8z$	$-8z^3$	$+32z^2$	$-160z$
$+65$	$+65z^2$	$-260z$	$+1300$

And

$z^2 - 4z + 20 = 0$ has solutions $z = 2 - 4i$ and $z = 2 + 4i$

$\therefore$ the roots of $z^4 - 12z^3 + 117z^2 - 420z + 1300 = 0$ are $4 - 7i$, $4 + 7i$, $2 - 4i$ and $2 + 4i$



Trapezium area $= \frac{1}{2}(8 + 14) \times 2 = 22$ units^2

41)

One root must be real.

Let $\mathrm{f}(z) = z^3 - 12z^2 + 47z - 54$

$$\mathrm{f}(1) = 1 - 12 + 47 - 54 \neq 0$$
$$\mathrm{f}(2) = 8 - 48 + 94 - 54 = 0 \ \therefore (z-2) \text{ is a factor}$$
$$(z-2)(z^2 - 10z + 27) = 0$$

So the roots are $z = 2$, $z = 5 + \sqrt{2}i$ and $z = 5 - \sqrt{2}i$

Area is $\frac{1}{2} \times 3 \times 2\sqrt{2} = 3\sqrt{2}$ units^2.

42 a)

Area is 64 units^2

42 b)

Two of the roots are $3 + 4i$ and $3 - 4i$

The other roots to form the square are $-5 + 4i$ and $-5 - 4i$

So the polynomial is:

$$\begin{aligned}
&(z - (3+4i))(z - (3-4i))(z - (-5+4i))(z - (-5-4i)) \\
&= (z^2 - 6z + 25)(z^2 + 10z + 41) \\
&= z^4 + 10z^3 + 41z^2 - 6z^3 - 60z^2 - 246z + 25z^2 + 250z + 1025 \\
&= z^4 + 4z^3 + 6z^2 + 4z + 1025
\end{aligned}$$

So $a = 4$, $b = 6$, $c = 4$ and $d = 1025$

43)

Let $\mathrm{f}(z) = z^3 - 7z^2 + az + b$

Let the complex roots be $p + qi$ and $p - qi$

$$\begin{aligned}
\mathrm{f}(z) &= (z-3)(z - (p+qi))(z - (p-qi)) \\
&= (z-3)(z^2 - 2pz + p^2 + q^2) \\
&= z^3 - 2pz^2 + (p^2+q^2)z - 3z^2 + 6pz - 3(p^2+q^2) \\
&= z^3 - (2p+3)z^2 + (p^2 + 6p + q^2)z - 3(p^2+q^2)
\end{aligned}$$

So $2p + 3 = 7 \ \therefore p = 2$

Area is $56 = \frac{1}{2} \times (2-3)(2q)$ which gives $q = -56$

$p = 2$ and $q = -56$

$$\begin{aligned}
\therefore z^3 - (4+3)z^2 + (4 + 12 + 3136)z - 3(4 + 3136) \\
= z^3 - 7z^2 + 3152z - 9420
\end{aligned}$$

$\therefore a = 3152$ and $b = -9420$

44)

$$\begin{aligned}
(z - (a+bi))(z - (a-bi))(z-k) &= (z^2 - 2az + a^2 + b^2)(z-k) \\
&= z^3 - 2az^2 + (a^2+b^2)z - kz^2 + 2kaz - k(a^2+b^2) \\
&= z^3 - (2a+k)z^2 + (a^2 + 2ka + b^2)z - k(a^2+b^2) \\
\mathrm{f}(z) &= z^3 - (2a+k)z^2 + (a^2 + 2ka + b^2)z - k(a^2+b^2)
\end{aligned}$$

45)

One of the roots must be real.

Let $\mathrm{f}(x) = z^3 - 23z^2 + 299z - 2197$

By testing values of x which are factors of -2197:

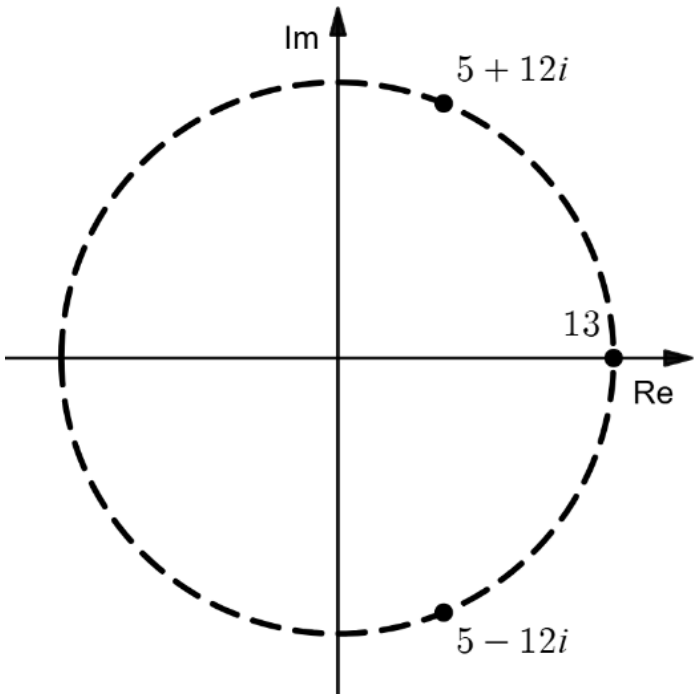
$$\mathrm{f}(13) = z^3 - 23z^2 + 299z - 2197 = 2197 - 3887 + 3887 - 2197 = 0$$

$\therefore (z - 13)$ is a factor of $z^3 - 23z^2 + 299z - 2197$

	z^2	$-10z$	$+169$
z	z^3	$-10z^2$	$+169z$
-13	$-13z^2$	$+130z$	-2197

and

$z^2 - 10z + 169 = 0$ has solutions $z = 5 + 12i$ and $z = 5 - 12i$



Distance to each point from the origin:

$\sqrt{5^2 + 12^2} = 13$, $\sqrt{5^2 + (-12)^2} = 13$ and 13 so they all lie on a circle of radius 13 centred at the origin. So that $|z| = 13$

46)

Complex roots have a distance of 6 between them so they are of the form $p + 3i$ and $p - 3i$

$$5 = \sqrt{p^2 + 3^2} \Rightarrow p = 4$$

Area is $42 = \frac{1}{2} \times 6 \times h$ so $h = 14$

So possible roots are $4 + 3i$, $4 - 3i$ and 18 or $4 + 3i$, $4 - 3i$ and -10

So the equations are:

$$\begin{aligned}(z - (4 + 3i))(z - (4 - 3i))(z - 18) &= (z^2 - 8z + 25)(z - 18) \\ &= z^3 - 26z^2 + 169z - 450 \\ (z - (4 + 3i))(z - (4 - 3i))(z + 10) &= (z^2 - 8z + 25)(z + 10) \\ &= z^3 + 2z^2 - 55z + 250\end{aligned}$$

So the equations are $f(z) = z^3 - 26z^2 + 169z - 450$ or $f(z) = z^3 + 2z^2 - 55z + 250$

47)

Since $x(x^3 - x^2 + 8x + 10) = 0$

Let $f(x) = x^3 - x^2 + 8x + 10$

$$f(-1) = -1 - 1 - 8 + 10 = 0$$

So $x = -1$ is a factor

So $x(x + 1)(x^2 - 2x + 10) = 0$ has solutions 0, -1, $1 + 3i$ and $1 - 3i$

$\therefore a = -1$

So when $a = -1$, $-2i$ is a root of

$$z^4 - 3z^2 + 14z - 28 = 0$$

Since $(z - 2i)(z + 2i) = z^2 + 4$

$$z^4 - 3z^2 - 28 = (z^2 + 4)(z^2 - 7)$$

$\therefore$ the values of z are $z = 2i$, $z = -2i$, $z = \sqrt{7}$ and $z = -\sqrt{7}$

48 a)

$$a + 3 - 3i$$

48 b)

$$\begin{aligned}&(z - b)(z - a)(z - (a + 3 + 3i))(z - (a + 3 - 3i))\\ &= (z - b)(z - a)(z^2 - (2a + 6)z + a^2 + 6a + 18)\\ &= (z^2 - (a + b)z + ab)(z^2 - (2a + 6)z + a^2 + 6a + 18)\end{aligned}$$

Compare cubic coefficients:

$$-(a + b) - (2a + 6) = -8 \Rightarrow b + 3a = 2 \Rightarrow b = 2 - 3a$$

Compare the quadratic coefficients:

$$(a + b)(2a + 6) + ab + (a^2 + 6a + 18) = 6$$

Substitute in $b = 2 - 3a$ and solve:

$$\begin{aligned}(a + 2 - 3a)(2a + 6) + a(2 - 3a) + (a^2 + 6a + 18) &= 6\\ (2 - 2a)(2a + 6) + 2a - 3a^2 + a^2 + 6a + 18 &= 6\\ -4a^2 - 12a + 4a + 12 + 2a - 3a^2 + a^2 + 6a + 18 &= 6\\ -6a^2 + 24 &= 0\\ a^2 &= 4\end{aligned}$$

$\therefore a = 2$ or $a = -2$ But since $a > 0$

$a = 2$

$a = 2$ gives $b = -4$

$\therefore a = 2$ and $b = -4$

48 c)

$\therefore$ the solutions are $z = 2$, $z = -4$, $z = 5 - 3i$ and $z = 5 + 3i$

49 a)

Two of the solutions are $5 + i$ and $5 - i$

$$(z - (5 + i))(z - (5 - i)) = z^2 - 10z + 26$$

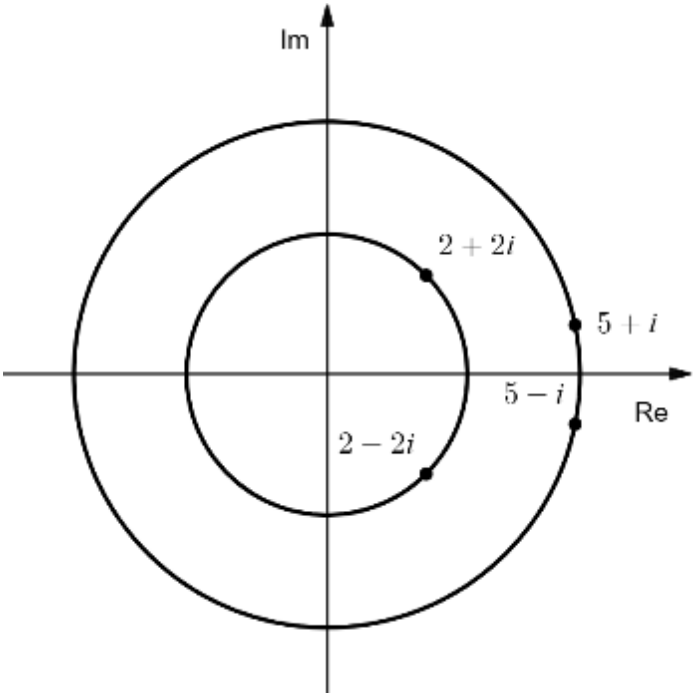
	z^2	$-4z$	$+8$
z^2	z^4	$-4z^3$	$+8z^2$
$-10z$	$-10z^3$	$+40z^2$	$-80z$
$+26$	$+26z^2$	$-104z$	$+208$

And

$z^2 - 4z + 8 = 0$ has solutions $z = 2 - 2i$ and $z = 2 + 2i$

$\therefore$ the roots of $z^4 - 14z^3 + 74z^2 - 184z + 208 = 0$ are $5 + i$, $5 - i$, $2 - 2i$ and $2 + 2i$

49 b)



Radius of larger circle: $\sqrt{5^2 + 1^2} = \sqrt{26}$

Radius of smaller circle: $\sqrt{2^2 + 2^2} = \sqrt{8}$

Area is $\pi \times 26 - \pi \times 8 = 18\pi$ units^2

50 a)

$z^2 - 1 = 0$ has roots $z = 1$ and $z = -1$

$$1 - 1 = 0$$

50 b)

$$z^3 - 1 = 0$$
$$(z-1)(z^2+z+1) = 0$$

has roots $z = 1$, $z = -\frac{1}{2} + \frac{\sqrt{3}}{2}i$ and $z = -\frac{1}{2} - \frac{\sqrt{3}}{2}i$

$$1 - \frac{1}{2} + \frac{\sqrt{3}}{2}i - \frac{1}{2} - \frac{\sqrt{3}}{2}i = 0$$

50 c)

$$z^4 - 1 = 0$$
$$(z^2-1)(z^2+1) = 0$$

has roots $z = 1$, $z = -1$, $z = i$ and $z = -i$

$$1 - 1 + i - i = 0$$

51 a)

Since $z = p + qi$ and $z = p - qi$ is a root:

$$\begin{aligned} pz^3 + qz^2 + qz + p &= (z-k)(p+qi)(p-qi) \\ &= (z^2 - 2pz + p^2 + q^2)(z-k) \\ &= z^3 - 2pz^2 + (p^2+q^2)z - kz^2 + 2kpz - k(p^2+q^2) \\ &= z^3 - (2p+k)z^2 + (p^2 + 2kp + q^2)z - k(p^2+q^2) \end{aligned}$$

Since the linear term is equal to the quadratic term:

$$\begin{aligned} p^2 + 2kp + q^2 &= -(2p+k) \\ p^2 + q^2 + 2p &= -2kp - k \\ p^2 + q^2 + 2p &= -k(2p+1) \end{aligned}$$

So $k = -\frac{p^2+q^2+2p}{2p+1}$ as req.

51 b)

$q = 0$ so

$$k = -\frac{p^2+2p}{2p+1} = -\frac{p(p+2)}{2p+1}$$

52)

Assume that it has two complex roots, $p + qi$ and $p - qi$

So that

$$(z-(p+qi))(z-(p-qi)) = z^2 - 2pz + p^2 + q^2$$

Assume a real root exists, k so that

$$\begin{aligned} \mathrm{f}(z) &= (z-(p+qi))(z-(p-qi))(z-k) \\ &= (z^2 - 2pz + p^2 + q^2)(z-k) \\ &= z^3 - 2pz^2 + (p^2+q^2)z - kz^2 + 2pkz - k(p^2+q^2) \\ &= z^3 - (2p+k)z^2 + (p^2 + 2pk + q^2)z - k(p^2+q^2) \end{aligned}$$

Since

$$p^2 + 2pk + q^2 = 0$$

For p to be real then the discriminant of $p^2 + 2pk + q^2$ must be ≥ 0

$$\begin{aligned} \therefore (2k)^2 - 4q^2 &\geq 0 \\ 4k^2 - 4q^2 &\geq 0 \\ k^2 &\geq q^2 \end{aligned}$$

53)

Assume that it has two complex roots, $p + qi$ and $p - qi$

So that

$$(z - (p + qi))(z - (p - qi)) = z^2 - 2pz + p^2 + q^2$$

Assume a two real roots exist, m and n so that

$$\begin{aligned} \mathrm{f}(z) &= (z - (p + qi))(z - (p - qi))(z - m)(z - n) \\ &= (z^2 - 2pz + p^2 + q^2)(z^2 - (m + n)z + mn) \\ &= z^4 - (m + n)z^3 + mnz^2 - 2pz^3 + 2p(m + n)z^2 - 2pmnz + (p^2 + q^2)z^2 \\ &\quad - (p^2 + q^2)(m + n)z + mn(p^2 + q^2) \\ &= z^4 - (m + n + 2p)z^3 + (2p(m + n) + mn + (p^2 + q^2))z^2 \\ &\quad - (2pmn + (p^2 + q^2)(m + n))z + mn(p^2 + q^2) \end{aligned}$$

Since

$$\begin{aligned} 2pmn + (p^2 + q^2)(m + n) &= 0 \\ (m + n)p^2 + 2mnp + (m + n)q^2 &= 0 \end{aligned}$$

For p to be real then the discriminant of $(m + n)p^2 + 2mnp + (m + n)q^2$ must be ≥ 0

$$\begin{aligned} \therefore (2mn)^2 - 4(m + n)(m + n)q^2 &\geq 0 \\ 4(mn)^2 &\geq 4(m + n)^2 q^2 \\ (mn)^2 &\geq (m + n)^2 q^2 \\ \frac{(mn)^2}{(m + n)^2} &\geq q^2 \end{aligned}$$

Since m, n and q are real and non-zero then

$$\frac{mn}{m + n} \geq q \text{ as req.}$$